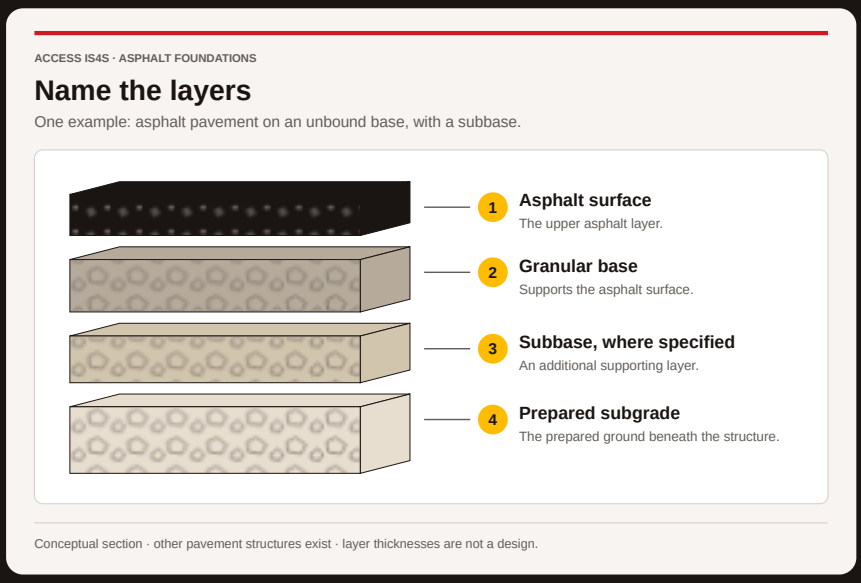
ACCESS · INDIGENOUS SKILLS 4 SUCCESS

Road geometry and acceptance

Volume 4 · F1, F2, F3, G1, G2, G3, G4, G5, G6

Classroom development edition · 30 October 2026



Asphalt Institute. (2007). The asphalt handbook (7th ed., MS-4). Original instructional adaptation by ACCESS IS4S (2026). Selected the unbound-base example, separated its layers and added functional descriptions. Other structures and design dimensions are not implied. Source locations: [{"figure": "1.6", "printed_page": 14, "pdf_page": 48}]. Original contributors: No separate contributor listed for this figure.

In this volume

1. How to use the volume and key terms
2. Drainage, grade and base
3. Verify slopes, crowns and grades
4. Final grade preparation
5. Gradation
6. Oil content
7. Application
8. Segregation
9. Density
10. Smoothness
11. Check the grade and the quantity

Use this volume

This volume provides complete classroom readings, worked examples, named practice tasks and answer guidance for road geometry and acceptance. Follow the learning route in each chapter, study the labelled figures and explain your reasoning. The separate answer book contains the worked keys and observation records.

Care for people, place and the next person using the work runs through the cases. Local First Nations, Métis and Inuit learners and partners bring distinct knowledge and priorities. Participation does not require sharing personal stories or cultural teachings, and no diagram claims to represent a Nation's protocol.

> Classroom authoring is complete for the listed competencies. Qualified technical review, actual fleet/site matching, a local teaching pilot and observed practical qualification are separate requirements. The practical records are ready for an instructor to configure and use; they do not certify unobserved work.

Read the picture, then test the idea

1. Identify the object, process or quantity shown.
2. Read labels, units and the figure's stated limits.
3. Connect the image to the chapter explanation.
4. Try the exercise and check against evidence.
5. Explain the result to another person and revise if needed.

Use AI as a checked working tool

Claude and ChatGPT can help explain provided documents and create supported files. Claude Code, Cowork and Codex can help with permitted multi-file tasks and repeatable workflows when their tools are available. Give a clear purpose, named sources, output formats, limits and checks. Ask for a findings table with page/cell evidence; verify it against the source. Reopen DOCX, XLSX, HTML or PDF outputs. For study audio/video, verify the script, captions, diagrams and actual playback. A generated result is not a technical approval.

Keep the term with its meaning

AGGREGATE
Mineral particles forming the solid skeleton of the mixture or granular layer.

BINDER
The asphalt material that coats and binds the aggregate in the specified mixture.

COMPACTED THICKNESS

Layer thickness after the required compaction process; distinct from loose depth behind the screed.

DENSITY REFERENCE
The specified comparison value used as the denominator of a percent-density calculation.

GRADATION
Distribution of aggregate particle sizes, commonly expressed as cumulative percent passing specified sieves.

HEAD OF MATERIAL
The quantity/distribution of mix immediately in front of the screed.

JOB MIX FORMULA (JMF)
The accepted mixture proportions and properties under the applicable quality/contract process.

LOCKOUT
Isolation, securing and verification of hazardous energy for work requiring protection.

LOT / SUB-LOT
Defined portions of work used for sampling, testing and acceptance under the contract.

OMM
Operation and maintenance manual matched to the actual machine, serial range and configuration.

RAP
Reclaimed asphalt pavement: recovered material containing aggregate and aged binder.

SDS
Safety data sheet for the actual product; read with its label and workplace procedure.

SCREED
The paver assembly that forms the mat; its response depends on setup, material and movement.

STATION / OFFSET
Location along an alignment / distance and direction from a defined reference.

TCP
Traffic control person with the required training and assigned role.

TRACEABILITY
A record connecting material, location, time, sample, result and authorized decision.

SVG
Scalable vector graphics: useful for clear diagrams whose labels and geometry must be checked.

DOCX / XLSX
Editable document / spreadsheet workbook; verify actual content and formulas after creation.

PDF / HTML
Page-based reading copy / browser document; inspect completeness, access and linked resources.

MP3 / MP4 / VTT

Audio file / video container / timed web captions; test playback and retain a transcript.

Study chapter · Outline F1

Drainage, grade and base

A pavement is a layered structure that must carry loads and manage water. A smooth new surface will not correct an unstable foundation or a blocked drainage route. Reading the whole road helps the crew protect access to homes, services and community spaces.

- Proper road drainage
- Sub grade
- Sub-base
- Base

LEARNING STEP 1

Proper road drainage

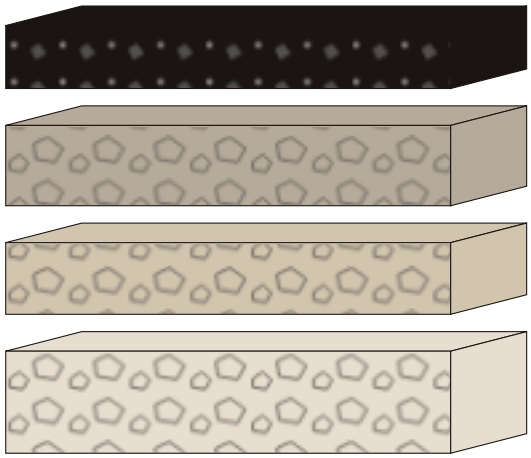
Surface water follows crossfall, longitudinal grade and low points toward an approved collection route. Subsurface water needs its own drainage path. Inspect ditches, catch basins, culverts and outlets against the drawings; a visible inlet is not proof of a functioning outlet. Sediment, fuel and uncured materials must be kept out of drainage systems. Do not redirect water toward neighbouring land or a watercourse because that route looks convenient. Report ponding, erosion and blocked drainage before covering the evidence. Approved environmental controls and local access commitments belong in the work plan.

Name the layers

ACCESS IS4S · ASPHALT FOUNDATIONS

Name the layers

One example: asphalt pavement on an unbound base, with a subbase.



1. **Asphalt surface**
   The upper asphalt layer.
2. **Granular base**
   Supports the asphalt surface.
3. **Subbase, where specified**
   An additional supporting layer.
4. **Prepared subgrade**
   The prepared ground beneath the structure.

Conceptual section · other pavement structures exist · layer thicknesses are not a design.

Note. Asphalt Institute. (2007). The asphalt handbook (7th ed., MS-4). Original instructional adaptation by ACCESS IS4S (2026). Selected the unbound-base example, separated its layers and added functional descriptions. Other structures and design dimensions are not implied. Source locations: [{"figure": "1.6", "printed_page": 14, "pdf_page": 48}]. Original contributors: No separate contributor listed for this figure. Full source context.

LEARNING STEP 2

Sub grade

Subgrade is the prepared soil foundation beneath the pavement structure. Soil type, moisture, density, frost susceptibility and soft areas influence performance. Rutting under equipment, pumping water and inconsistent support require evaluation; adding asphalt is not a substitute for correcting the cause. Geotextiles can separate materials or provide filtration, depending on the specified product and design. Geogrids have different functions. Check product identity, overlap, orientation and protection requirements before placement. Damaged or wrongly installed fabric cannot be assumed effective because it is hidden.

LEARNING STEP 3

Sub-base

Sub-base distributes loading and can provide a working platform or drainage function. Its specified grading, thickness, moisture and compaction requirements govern placement. Approved recycled material may be suitable, but availability alone does not establish compliance. Place in controlled lifts, protect against

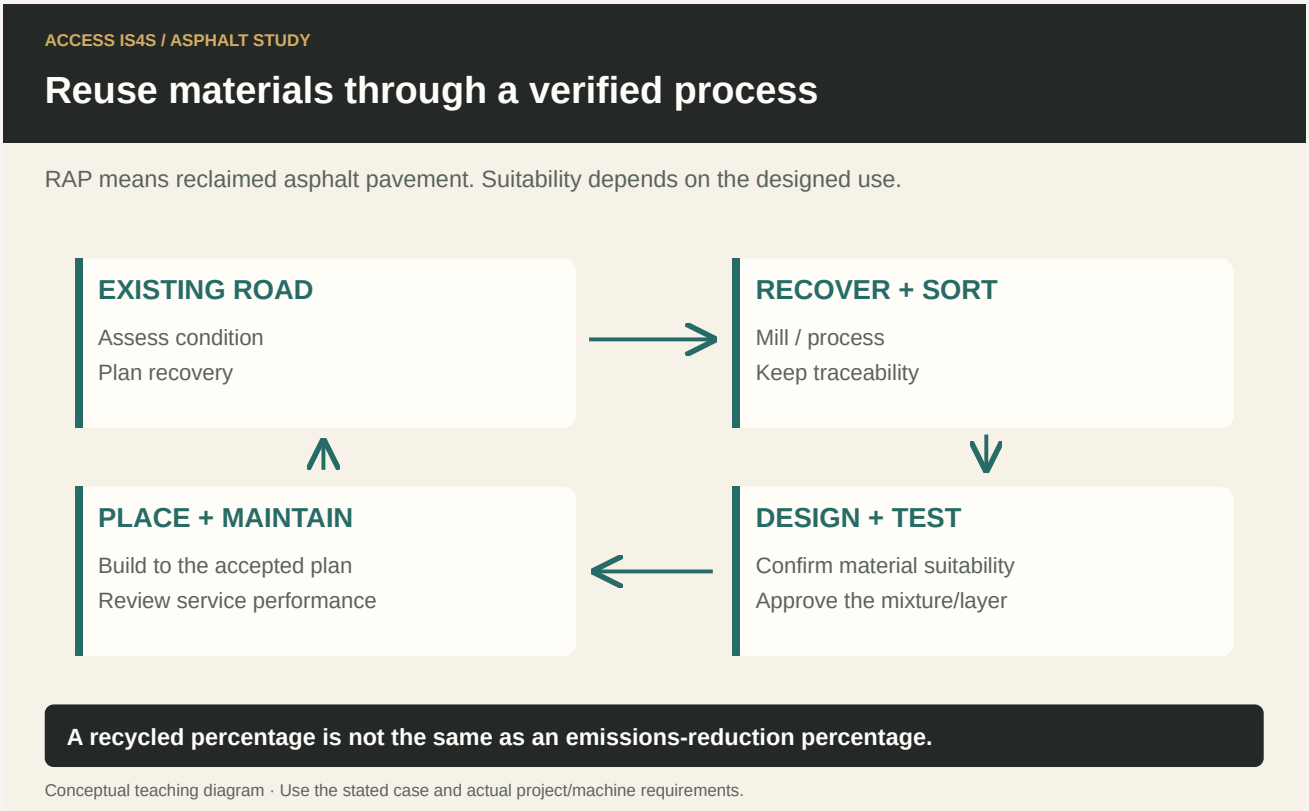
contamination and check each accepted layer before burying it. Segregation, excessive fines and trapped water can change how the layer behaves. Record the material source and test results so a later concern can be traced to an actual location and lift.

LEARNING STEP 4

Base

Base is the prepared layer directly supporting the asphalt in a typical flexible pavement. Crushed aggregate from a quarry or pit, approved dredged or river-derived material, and recycled products each require the project's source authorization and material acceptance. Their names do not establish equal performance or permission to extract them. Check surface elevation, crossfall, density, cleanliness and drainage. A thin low spot in the base changes asphalt demand and may create a local thickness problem. Maintain the accepted base until paving starts; traffic or rain can alter it after inspection.

Reuse materials through a verified process



Note. ACCESS IS4S. (2026). Reuse materials through a verified process [Original instructional SVG]. Conceptual explanation based on the cited chapter readings; geometry and operating arrangements are illustrative, not prescribed settings.

Worked example · Follow the water

A 120 m access road has a sound-looking base. A blocked catch basin sits at its low point, and water is pumping through one wheel path. The paving crew is ready.

Think it through.

Hold the affected work and identify the drainage and support problems. Protect public access and the inlet, document the location, and request the responsible person's correction and acceptance. Do not treat extra asphalt as a drainage repair. Record the repaired outlet and accepted foundation before release.

Try it and explain your thinking

Draw the four typical layers from prepared soil to asphalt

Draw the four typical layers from prepared soil to asphalt. Label one check at each interface.

Skills you use: Reading, Writing, Creativity and innovation.

An approved 80 m × 4 m sub-base lift is 0.15 m compacted thickness

An approved 80 m × 4 m sub-base lift is 0.15 m compacted thickness. Find its compacted volume. Can you order that many loose cubic metres without further information?

Skills you use: Numeracy, Problem solving.

A partner proposes opening a ditch toward a nearby creek to remove ponded water

A partner proposes opening a ditch toward a nearby creek to remove ponded water. Give a respectful response and a next step.

Skills you use: Communication, Collaboration, Adaptability.

Show the skill in supervised practice

Machine/model/serial or classroom case: _____ · Manual/procedure and revision: _____ · Date/location: _____ · Learner/observer: _____

Record each criterion as 0—not yet demonstrated, 1—with task-solving support, or 2—independently under the stated conditions. Mark unobserved criteria separately; do not convert missing opportunity into a zero. The assessor determines the required repeat practice and applicable qualification requirements. A critical failure means hold, correction and reassessment regardless of the numerical result. Use safe simulations for classroom reasoning; actual operation requires the selected equipment, approved conditions and competent supervision.

What the observer checks	Result	Evidence
Critical: Safe access, surroundings and communication maintained.	Pass / hold	Observation: _____
Critical: Required energy, heat, movement and public-access controls maintained for the actual task.	Pass / hold	Observation: _____
Identify actual layer boundaries on an approved section.	0 / 1 / 2	Evidence/help: _____
Trace a drainage route from surface to authorized outlet.	0 / 1 / 2	Evidence/help: _____
Record and communicate a foundation defect before covering it.	0 / 1 / 2	Evidence/help: _____

Actual conditions and support: _____

Feedback, next practice and authorized release/status: _____

Sources and further reading

- SkilledTradesBC. (2016). *Asphalt Paving/Laydown Technician program outline*, F1, printed p. 48; source PDF pp. 50–50.
- Asphalt Institute. (2007). The asphalt handbook (7th ed., MS-4), Chapter 5: Soils and Drainage for Asphalt Pavements; source PDF pp. 281–314. Open publication.
- Asphalt Institute. (2007). The asphalt handbook (7th ed., MS-4), Chapter 6: Thickness Design; source PDF pp. 315–340. Open publication.
- BC Ministry of Transportation and Transit. (2025). 2025 Standard Specifications for Highway Construction, Volume 1. Open publication.

Figure-specific pages, editions and original contributors are credited beside each image. Current project requirements and the matching machine instructions determine practical use.

Study chapter · Outline F2

Verify slopes, crowns and grades

Grade work turns a drawing into reliable elevations on the ground. Keep the benchmark, units, drawing revision and sign convention visible in every calculation. Another person should be able to repeat your result without guessing what a number means.

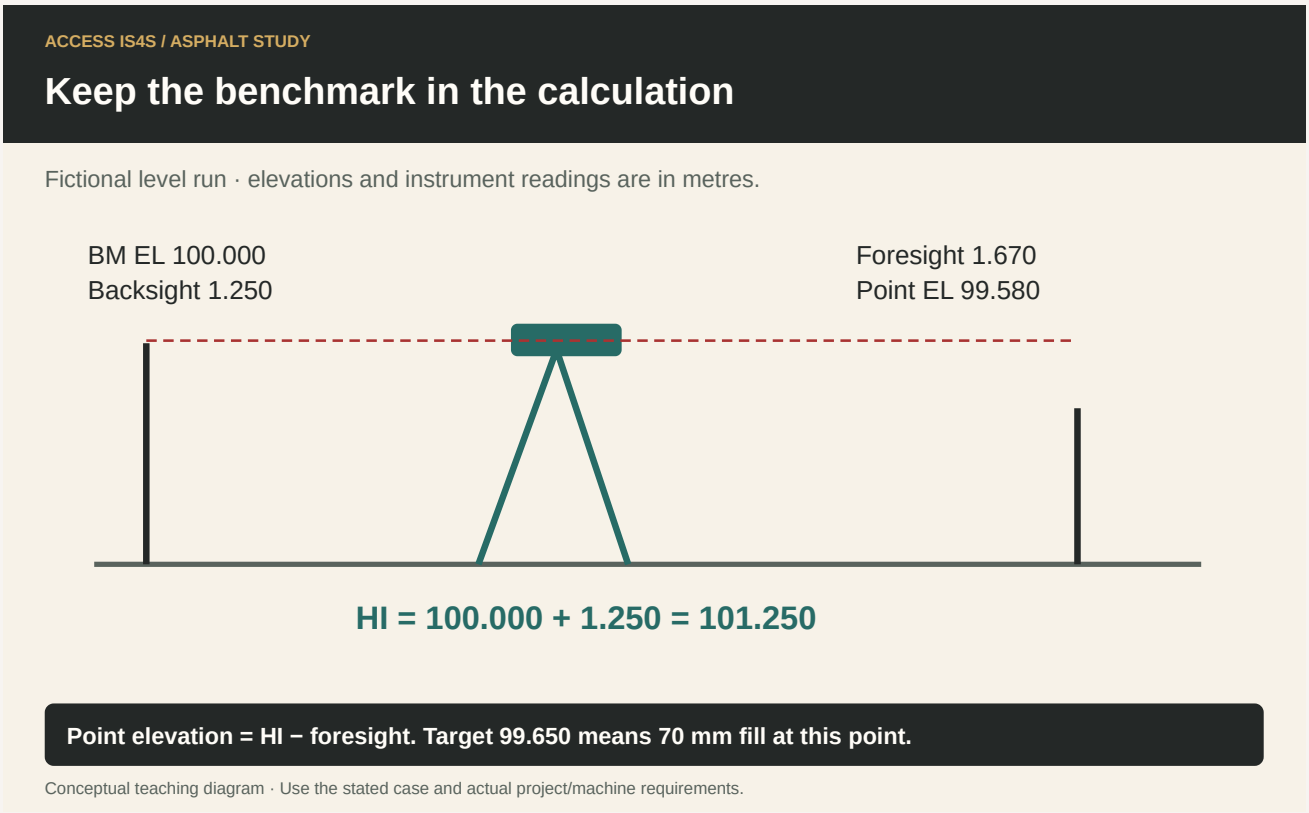
- Confirm plans/blueprints
- Define role of Surveyor
- Define role of Grade Person
- Typical calculations

LEARNING STEP 1

Confirm plans/blueprints

Read the title block, north arrow, legend, scale, revision and detail references before measuring. A plan is the overhead view; a profile follows an alignment; a cross-section cuts across it. Profiles often exaggerate the vertical scale, so the drawn angle is not the road’s actual slope. Use stated dimensions before scaling, especially on resized screens or prints. One inch is exactly 25.4 mm and one foot exactly 0.3048 m. Horizontal run is used for ordinary grade calculations; a tape stretched along a slope measures a longer distance.

Keep the benchmark in the calculation



Note. ACCESS IS4S. (2026). Keep the benchmark in the calculation [Original instructional SVG]. Conceptual explanation based on the cited chapter readings; geometry and operating arrangements are illustrative, not prescribed settings.

LEARNING STEP 2

Define role of Surveyor

The surveyor establishes or verifies control and supplies elevations, alignment and construction information within their assigned role. Cut and fill at a profile station normally refer to the defined centreline, not automatically to the entire cross-section. Pipe invert is the inside-bottom elevation. Depth to invert and cover above the pipe crown are different quantities; calculating cover requires the appropriate pipe dimensions. Quantity work combines accepted geometry with a justified allowance for settlement or compaction. A material list includes identity, size, quantity, units and drawing reference, not just an unexplained total.

LEARNING STEP 3

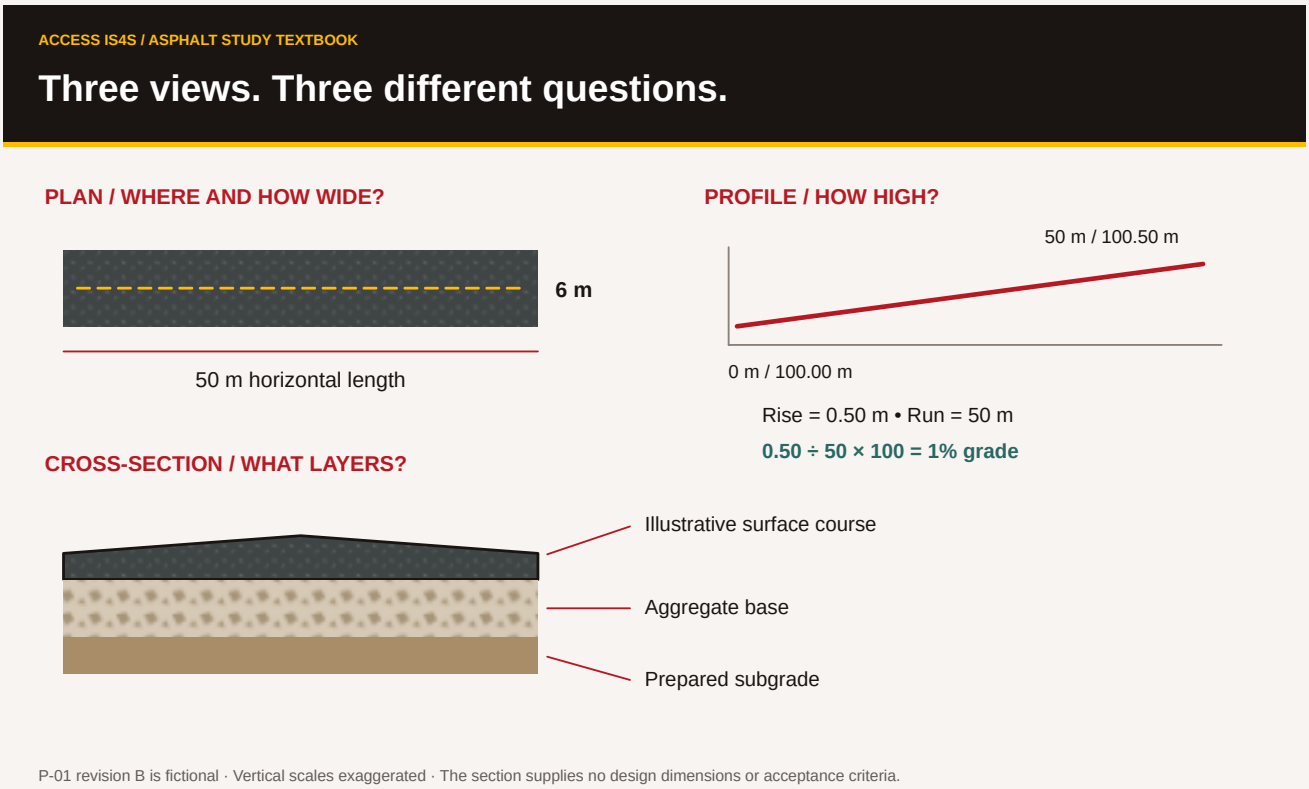
Define role of Grade Person

The grade person transfers authorized control to usable offset stakes, marks and machine references. Protect the original control and record offsets so a stake displaced from the construction line is not mistaken for the line itself. Confirm benchmark identity, datum, instrument setup and backsight. If marks conflict or a stake moves, stop using it and obtain verification. Do not average conflicting elevations. Read back instructions with station, side, offset, target elevation and cut/fill direction.

Typical calculations

For differential levelling, height of instrument (HI) = known elevation + backsight; point elevation = HI − foresight. For grade, rise/run × 100 gives percent when both use the same units. A 2% crossfall across a 3.5 m horizontal half-width gives 0.070 m or 70 mm drop. Use a spirit level, builder's level, laser and automatic slope control only with their setup, checking and operating instructions. Check calibration and an independent reference. Instrument displays do not make an incorrect benchmark correct. Record tolerances from the project rather than inventing a universal pass band.

Three views. Three different questions.



Note. ACCESS IS4S. (2026). Three views. Three different questions. [Original instructional SVG]. Conceptual illustration; read the stated conditions in the lesson.

Worked example · A level-book entry

Benchmark EL 100.000 m; backsight 1.250 m; foresight to prepared base 1.670 m. Target base EL 99.650 m.

Think it through.

HI = 101.250 m. Existing base = 101.250 − 1.670 = 99.580 m. Target minus existing = +0.070 m: fill 70 mm at this point, subject to the specified material and compaction method. Check against the correct drawing revision and benchmark before directing work.

Try it and explain your thinking

Use the same benchmark with backsight 1.400 m and foresight 1.600 m

Use the same benchmark with backsight 1.400 m and foresight 1.600 m. Target EL is 99.650 m. State cut or fill.

Skills you use: Numeracy, Reading.

A profile uses horizontal 1:500 and vertical 1:50

A profile uses horizontal 1:500 and vertical 1:50. Explain why measuring its apparent angle is unreliable.

Skills you use: Digital, Problem solving.

Write a handover for a disturbed offset stake at station 0+080, right side.

Write a handover for a disturbed offset stake at station 0+080, right side.

Skills you use: Writing, Communication, Collaboration.

Show the skill in supervised practice

Machine/model/serial or classroom case: _____ · Manual/procedure and revision: _____ · Date/location: _____ · Learner/observer: _____

Record each criterion as 0—not yet demonstrated, 1—with task-solving support, or 2—independently under the stated conditions. Mark unobserved criteria separately; do not convert missing opportunity into a zero. The assessor determines the required repeat practice and applicable qualification requirements. A critical failure means hold, correction and reassessment regardless of the numerical result. Use safe simulations for classroom reasoning; actual operation requires the selected equipment, approved conditions and competent supervision.

What the observer checks	Result	Evidence
Critical: Safe access, surroundings and communication maintained.	Pass / hold	Observation: _____
Critical: Required energy, heat, movement and public-access controls maintained for the actual task.	Pass / hold	Observation: _____
Set up and check spirit, builder's and laser levels under supervision.	0 / 1 / 2	Evidence/help: _____
Verify automatic slope control against an independent approved reference.	0 / 1 / 2	Evidence/help: _____
Maintain specified grade and slope and retain a legible level-book record.	0 / 1 / 2	Evidence/help: _____

Actual conditions and support: _____

Feedback, next practice and authorized release/status: _____

Sources and further reading

- SkilledTradesBC. (2016). *Asphalt Paving/Laydown Technician program outline*, F2, printed p. 49; source PDF pp. 51–52.
- Leica Geosystems. (n.d.; manufacturer brochure). Leica NA700 Series Automatic Levels. Open publication.
- Leica Geosystems. (n.d.). Leica NA700 Series Automatic Levels: Technical Data. Open publication.
- Seymour Midwest. (n.d.). MP Metal Professional Measuring Wheels: Counter Alignment. Open publication.
- BC Ministry of Transportation and Transit. (2025). 2025 Standard Specifications for Highway Construction, Volume 1. Open publication.

Figure-specific pages, editions and original contributors are credited beside each image. Current project requirements and the matching machine instructions determine practical use.

Study chapter · Outline F3

Final grade preparation

Final grade preparation is the last opportunity to correct the support and geometry before the mat covers them. Work from accepted levels, drainage and utility details. A neat edge and a clean base help the paving crew place the intended thickness.

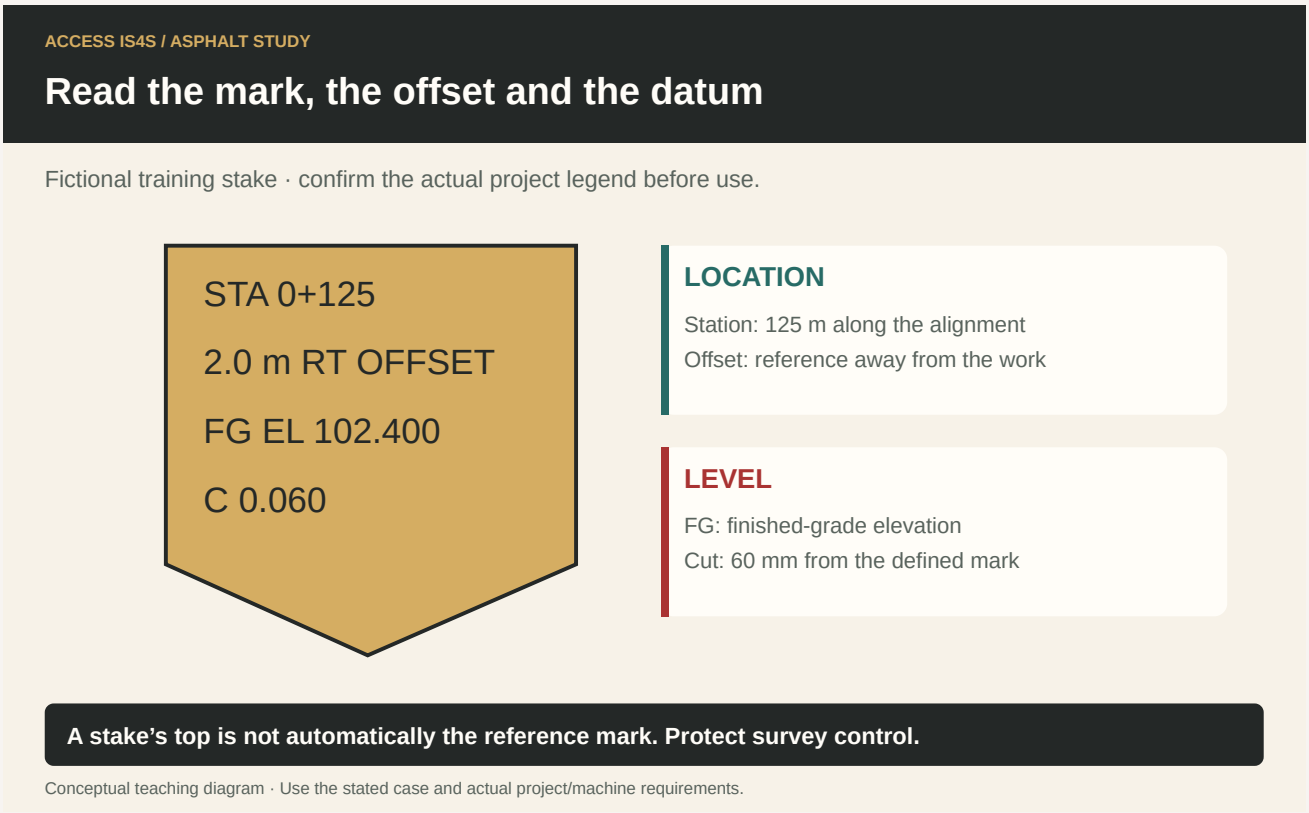
- Check and raise utilities
- Fine grading
- Boning crown/cross-fall
- Limit of asphalt layout

LEARNING STEP 1

Check and raise utilities

Locate utility structures using the project information and required locating process. Confirm casting type, ownership, target elevation and approved adjustment detail. Raising a frame may require rings, approved bedding, structural support and curing time; it is not simply lifting the lid. Protect openings, coordinate access and traffic control, and prevent loose material entering the system. Only authorized workers make the adjustment. Recheck finished elevation and crossfall around the structure so the paving joint does not create a pond or abrupt bump.

Read the mark, the offset and the datum



Note. ACCESS IS4S. (2026). Read the mark, the offset and the datum [Original instructional SVG]. Conceptual explanation based on the cited chapter readings; geometry and operating arrangements are illustrative, not prescribed settings.

LEARNING STEP 2

Fine grading

Fine grading removes high spots and fills low areas with the approved material and method. Check moisture and compaction as required, remove contamination and protect the surface from rutting. Do not conceal soft spots with a thin scatter of dry aggregate. Confirm that repairs are accepted and that the base is ready for the specified prime or tack treatment, where applicable. Record the inspection location and time because construction traffic and weather may change the surface after it was checked.

LEARNING STEP 3

Boning crown/cross-fall

Boning uses aligned sighting references to compare intermediate levels with a defined grade. All rods and reference heights must be consistent, and the method needs a verified starting and finishing reference. Check crown and crossfall with appropriate instruments and the drawing's actual geometry. A crown is not always centred, and curves, transitions, intersections and accessibility features may have different requirements. Check machine slope calibration against an independent measure. Agreement between two displays using the same wrong reference is not independent verification.

LEARNING STEP 4

Limit of asphalt layout

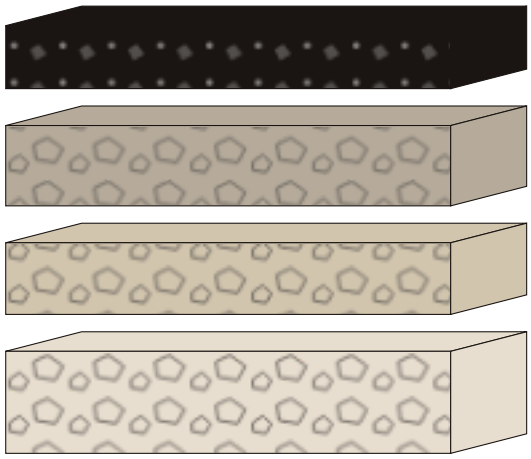
Mark the finished mat limits from approved alignment, width, edge and tie-in details. Include tapers, entrances, curb offsets, construction joints and the paving sequence. Distinguish the paver’s nominal screed width from the specified finished width and any working allowance. Measure at identified stations and record changes. Discuss public access and emergency routes before changing the work boundary. If the layout conflicts with a utility or existing surface, request a resolved detail before placing material.

Name the layers

ACCESS IS4S · ASPHALT FOUNDATIONS

Name the layers

One example: asphalt pavement on an unbound base, with a subbase.



1. **Asphalt surface** — The upper asphalt layer.
2. **Granular base** — Supports the asphalt surface.
3. **Subbase, where specified** — An additional supporting layer.
4. **Prepared subgrade** — The prepared ground beneath the structure.

Conceptual section · other pavement structures exist · layer thicknesses are not a design.

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Worked example · Width before tonnage

A rectangular paving area is 60 m long. The approved width is 4.2 m, but the marked limits measure 4.0 m.

Think it through.

Approved area = 252 m²; marked area = 240 m². The shortfall is 12 m². Verify the drawing and correct the layout before ordering or placing to the wrong boundary. Extra mix later does not automatically restore the intended edge support and joint quality.

Try it and explain your thinking

The target casting EL is 102.340 m and its verified current EL is 102.295 m

The target casting EL is 102.340 m and its verified current EL is 102.295 m. Find the required elevation change.

Skills you use: Numeracy, Problem solving.

Build a five-item release record for the base.

Build a five-item release record for the base.

Skills you use: Writing, Digital.

A worker suggests smoothing a pumping area and paving immediately to keep the delivery slot

A worker suggests smoothing a pumping area and paving immediately to keep the delivery slot. Respond.

Skills you use: Communication, Adaptability, Collaboration.

Show the skill in supervised practice

Machine/model/serial or classroom case: _____ · Manual/procedure and revision: _____ · Date/location: _____ · Learner/observer: _____

Record each criterion as 0—not yet demonstrated, 1—with task-solving support, or 2—independently under the stated conditions. Mark unobserved criteria separately; do not convert missing opportunity into a zero. The assessor determines the required repeat practice and applicable qualification requirements. A critical failure means hold, correction and reassessment regardless of the numerical result. Use safe simulations for classroom reasoning; actual operation requires the selected equipment, approved conditions and competent supervision.

What the observer checks	Result	Evidence
Critical: Safe access, surroundings and communication maintained.	Pass / hold	Observation: _____
Critical: Required energy, heat, movement and public-access controls maintained for the actual task.	Pass / hold	Observation: _____
Check utility elevations and protected openings with the responsible person.	0 / 1 / 2	Evidence/help: _____
Verify crown, crossfall and width against the current layout.	0 / 1 / 2	Evidence/help: _____
Complete a release record identifying unresolved defects.	0 / 1 / 2	Evidence/help: _____

Actual conditions and support: _____

Feedback, next practice and authorized release/status: _____

Sources and further reading

- SkilledTradesBC. (2016). *Asphalt Paving/Laydown Technician program outline*, F3, printed p. 51; source PDF pp. 53–53.
- Asphalt Institute. (2007). The asphalt handbook (7th ed., MS-4), Chapter 8: Hot Mix Asphalt Paving; source PDF pp. 391–446. Open publication.
- BC 1 Call. (August 2020). Safety Is a Shared Responsibility: Six Steps to Safe Digging. Open publication.
- BC Ministry of Transportation and Transit. (2025). 2025 Standard Specifications for Highway Construction, Volume 1. Open publication.

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Study chapter · Outline G1

Gradation

Quality control connects the planned pavement with evidence from production and placement. The crew needs to know which document governs, which tests apply and who can release or reject work. This chapter uses the supplied BC Ministry of Transportation DB 2006 Section 502 for a dated contract-reading exercise.

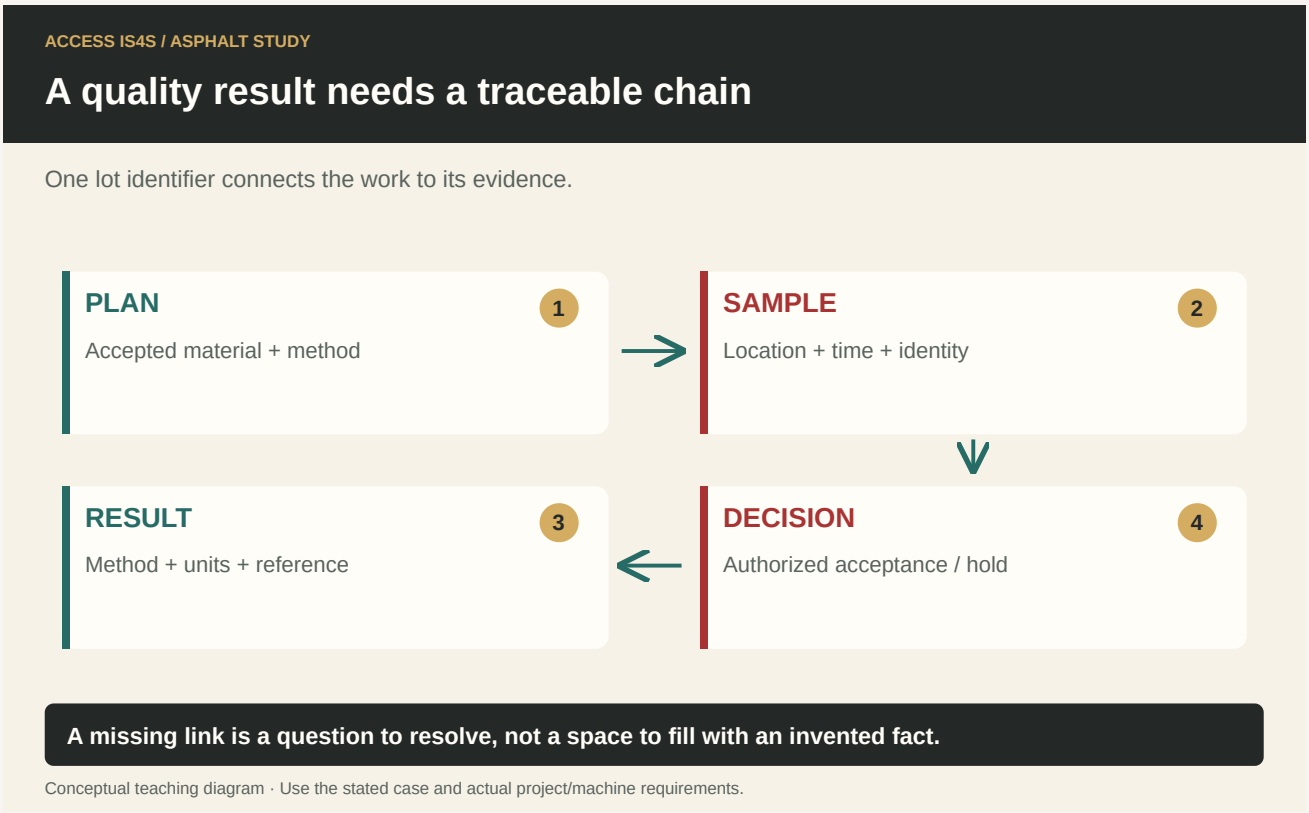
- MOT Specs
- The audit process
- Penalty/bonus
- Limit of asphalt layout

LEARNING STEP 1

MOT Specs

A quality management plan identifies materials, accepted mix design, sampling and testing, production controls, placement checks, records and responsibilities. Quality control is the producer/contractor’s process control; quality assurance provides the owner’s confidence and verification through the specified system. Neither term means that a crew can self-approve a departure. In the 2006 example, a mix design becomes the job mix formula (JMF) through the required written acceptance process. Identify the actual contract edition, special provisions and amendments before using any limit on a real job.

A quality result needs a traceable chain



Note. ACCESS IS4S. (2026). A quality result needs a traceable chain [Original instructional SVG]. Conceptual explanation based on the cited chapter readings; geometry and operating arrangements are illustrative, not prescribed settings.

LEARNING STEP 2

The audit process

An audit follows evidence: approved procedure, actual activity, dated record, result and corrective action. A missing record and a failed result are different findings, and both require a response. Trace a sample from station and lot through laboratory report to disposition. Check instrument identification and calibration records. Record what was observed without inventing motives. Close a finding only when the responsible authority accepts the evidence of correction. A tidy binder with untraceable sample numbers does not demonstrate control.

LEARNING STEP 3

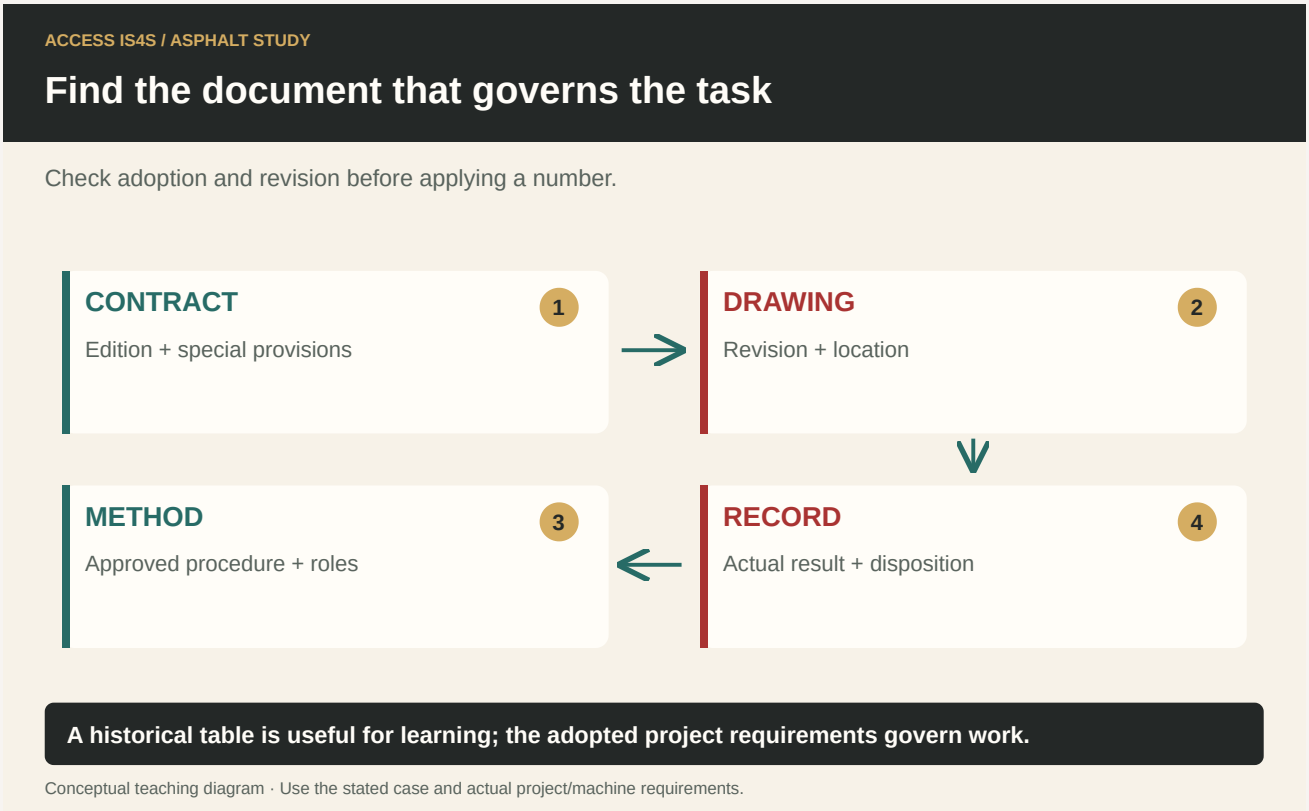
Penalty/bonus

Payment adjustments are contract calculations tied to defined lots, tests and acceptance rules. A bonus for one property does not cancel rejection under another. The 2006 tables separately address binder content, gradation, application rate, segregation, density and smoothness. Always state parameter, units, applicable lift, table and edition. Crowns and crossfalls remain geometric requirements even where a payment table does not express them. Learners calculate an isolated adjustment first, then check all other acceptance conditions before discussing overall payment.

Limit of asphalt layout

Define the paving limits, lift, station range, lane and measured area before assigning tickets or samples to a lot. Mark intersections, letdowns and other geometry according to the contract's measurement method. A layout change can affect quantity, sample locations and the meaning of a payment calculation. For the 2006 gradation exercise, Table 502-G compares departures from the JMF on specified sieves; it is not a width table. Keep layout verification and the appropriate quality-table lookup linked through the same lot identifier.

Find the document that governs the task



Note. ACCESS IS4S. (2026). Find the document that governs the task [Original instructional SVG]. Conceptual explanation based on the cited chapter readings; geometry and operating arrangements are illustrative, not prescribed settings.

Worked example · One good result is not full acceptance

A fictional 600 t lot qualifies for a +$0.75/t gradation adjustment under the complete applicable table conditions. A separate density result is under review.

Think it through.

The isolated gradation adjustment is 600 × $0.75 = +$450. Record it as provisional for this parameter. Do not declare the lot accepted or payable until all applicable requirements and the authorized disposition are resolved.

Try it and explain your thinking

An audit finds sample Q17 on a report but no matching location log

An audit finds sample Q17 on a report but no matching location log. Write a finding and action.

Skills you use: Writing, Problem solving.

For a fictional 420 t lot, calculate a –$1.50/t isolated adjustment.

For a fictional 420 t lot, calculate a –$1.50/t isolated adjustment.

Skills you use: Numeracy, Reading.

Compare the 2006 document and the current project document in a digital register

Compare the 2006 document and the current project document in a digital register. What fields prevent accidental substitution?

Skills you use: Digital, Collaboration, Adaptability.

Show the skill in supervised practice

Machine/model/serial or classroom case: _____ · Manual/procedure and revision: _____ · Date/location: _____ · Learner/observer: _____

Record each criterion as 0—not yet demonstrated, 1—with task-solving support, or 2—independently under the stated conditions. Mark unobserved criteria separately; do not convert missing opportunity into a zero. The assessor determines the required repeat practice and applicable qualification requirements. A critical failure means hold, correction and reassessment regardless of the numerical result. Use safe simulations for classroom reasoning; actual operation requires the selected equipment, approved conditions and competent supervision.

What the observer checks	Result	Evidence
Critical: Safe access, surroundings and communication maintained.	Pass / hold	Observation: _____
Critical: Required energy, heat, movement and public-access controls maintained for the actual task.	Pass / hold	Observation: _____
Trace one fictional lot through plan, sample, result and disposition.	0 / 1 / 2	Evidence/help: _____
Identify the governing edition and amendments.	0 / 1 / 2	Evidence/help: _____
Communicate a quality hold with exact location and evidence.	0 / 1 / 2	Evidence/help: _____

Actual conditions and support: _____

Feedback, next practice and authorized release/status: _____

Sources and further reading

- SkilledTradesBC. (2016). *Asphalt Paving/Laydown Technician program outline*, G1, printed p. 52; source PDF pp. 54–54.
- Asphalt Institute. (2007). The asphalt handbook (7th ed., MS-4), Chapter 3: Mineral Aggregates and Mixture Additives; source PDF pp. 143–192. Open publication.
- Asphalt Institute. (2007). The asphalt handbook (7th ed., MS-4), Chapter 10: Quality Control and Acceptance of Hot Mix Asphalt; source PDF pp. 479–552. Open publication.
- British Columbia Ministry of Transportation. (2006). Design build standard specifications, Section 502; PDF pp. 297–318. Historical contract exercise. Open publication.
- BC Ministry of Transportation and Transit. (2025). 2025 Standard Specifications for Highway Construction, Volume 1. Open publication.

Figure-specific pages, editions and original contributors are credited beside each image. Current project requirements and the matching machine instructions determine practical use.

Oil content

In asphalt quality work, “oil content” commonly refers to asphalt binder content. It does not mean engine oil. Binder content affects the mixture, but its reported percentage is meaningless unless the mass basis and test method are known.

> Define Section 502

LEARNING STEP 1

Define Section 502

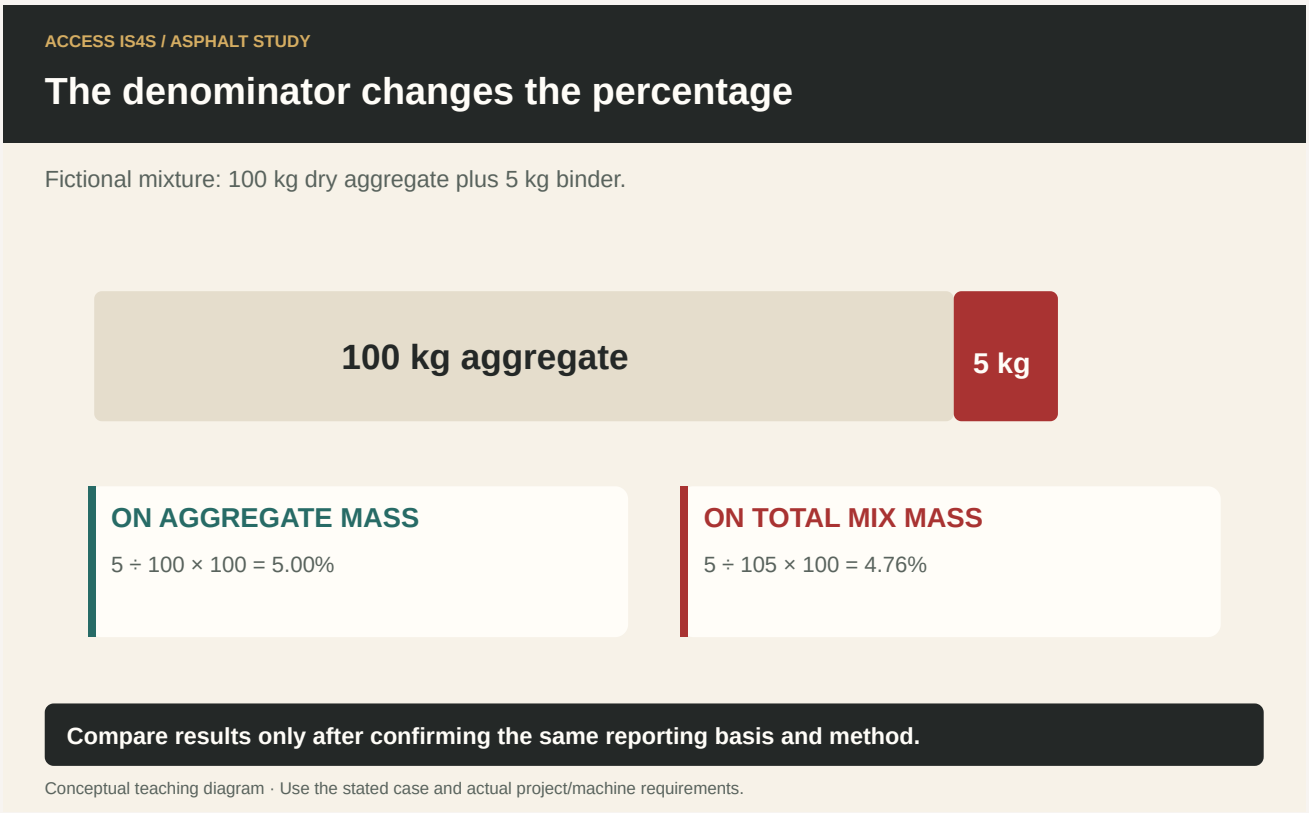
Section 502.12 of the supplied 2006 specification evaluates actual asphalt content against the approved content. Table 502-F expresses deviation in kilograms of binder per 100 kg of dry aggregate. This is a different basis from binder as a percentage of total mixture. For example, 5 kg binder with 100 kg dry aggregate is 5.00% on aggregate mass and 5/105 × 100 = 4.76% on total mixture mass. Never subtract results reported on different bases.

The historical procedure uses the sample mean from the specified sub-lot samples. Preserve sample identity, test method, lot and lift. Compute actual minus approved content, then use the magnitude of the deviation where the table applies to either above or below. Follow the contract’s rounding and reporting rules rather than rounding early to reach a favourable band.

In Table 502-F, a deviation of 0–0.15 earns +$0.50/t, 0.16–0.30 has no adjustment, and 0.31–0.35 gives –$1/t. Larger deviations have larger deductions and different top/lower-lift rejection boundaries. The full table and associated clauses govern; these are historical exercise values, not current project permission. A top-lift deviation greater than 0.50 triggers the stated rejection/remedial provisions. An operator does not fix a field result by independently changing the approved JMF.

Possible causes of an unexpected result include actual production variation, sampling problems, incorrect identities, test issues or a mistaken reporting basis. Preserve the original evidence and refer the issue to the quality lead. A single favourable retest is not automatic authority to discard an unfavourable result.

The denominator changes the percentage



Note. ACCESS IS4S. (2026). The denominator changes the percentage [Original instructional SVG]. Conceptual explanation based on the cited chapter readings; geometry and operating arrangements are illustrative, not prescribed settings.

2006 Table 502-F: binder-content adjustment

TABLE 502-F – PAYMENT ADJUSTMENTS FOR ASPHALT CONTENT

Deviation of the Actual Asphalt Content from the Approved Asphalt Content (either above or below) kg/100kg of Dry Aggregate	Payment Adjustment for Asphalt Content $ per tonne	
	Top Lift	Lower Lift
From 0 to 0.15	$0.50	$0.50
From 0.16 to 0.30	$0.00	$0.00
From 0.31 to 0.35	-$1.00	-$1.00
From 0.36 to 0.40	-$2.00	-$2.00
From 0.41 to 0.45	-$3.00	-$3.00
From 0.46 to 0.50	-$4.00	-$4.00
From 0.51 to 0.55	Reject	-$5.00
0.56 and greater	Reject	Reject
For top lift deviations of more than 0.50 the Contractor shall either overlay or remove and replace the Lot. For lower lift deviations of more than 0.55, no payment will be made and the Ministry Representative will determine whether removal and replacement is necessary.		

Note. British Columbia Ministry of Transportation. (2006). Design build standard specifications, Section 502 (section p. 12; PDF p. 308). Historical contract-reading example; complete clauses and actual project adoption govern. Full source context.

Worked example · Read the basis before the band

Approved binder content is 5.20 kg/100 kg dry aggregate. The three fictional sub-lot results are 5.45, 5.50 and 5.49 on that same basis. Lot mass is 500 t.

Think it through.

Mean = (5.45 + 5.50 + 5.49)/3 = 5.48. Absolute deviation = 0.28. Under the dated Table 502-F exercise, 0.16–0.30 gives $0/t, so this isolated adjustment is $0. This says nothing about other acceptance parameters.

Try it and explain your thinking

Approved content is 5.20 and the mean is 4.86, both kg/100 kg dry aggregate

Approved content is 5.20 and the mean is 4.86, both kg/100 kg dry aggregate. Find deviation and the historical isolated adjustment for 300 t.

Skills you use: Numeracy, Reading.

A report says 5.2% but has no basis

A report says 5.2% but has no basis. Can it be compared directly with 5.2 kg/100 kg dry aggregate?

Skills you use: Problem solving, Communication.

Write a two-line hold note for conflicting sample IDs.

Write a two-line hold note for conflicting sample IDs.

Skills you use: Writing, Collaboration.

Show the skill in supervised practice

Machine/model/serial or classroom case: _____ · Manual/procedure and revision: _____ · Date/location: _____ · Learner/observer: _____

Record each criterion as 0—not yet demonstrated, 1—with task-solving support, or 2—independently under the stated conditions. Mark unobserved criteria separately; do not convert missing opportunity into a zero. The assessor determines the required repeat practice and applicable qualification requirements. A critical failure means hold, correction and reassessment regardless of the numerical result. Use safe simulations for classroom reasoning; actual operation requires the selected equipment, approved conditions and competent supervision.

What the observer checks	Result	Evidence
Critical: Safe access, surroundings and communication maintained.	Pass / hold	Observation: _____

What the observer checks	Result	Evidence
Critical: Required energy, heat, movement and public-access controls maintained for the actual task.	Pass / hold	Observation: _____
Identify binder basis and lot/lift before calculation.	0 / 1 / 2	Evidence/help: _____
Read the complete applicable payment/rejection table.	0 / 1 / 2	Evidence/help: _____
Escalate a conflicting result without altering original evidence.	0 / 1 / 2	Evidence/help: _____

Actual conditions and support: _____

Feedback, next practice and authorized release/status: _____

Sources and further reading

- SkilledTradesBC. (2016). *Asphalt Paving/Laydown Technician program outline*, G2, printed p. 53; source PDF pp. 55–55.
- Asphalt Institute. (2007). The asphalt handbook (7th ed., MS-4), Chapter 2: Petroleum Asphalt; source PDF pp. 55–142. Open publication.
- Asphalt Institute. (2007). The asphalt handbook (7th ed., MS-4), Chapter 4: Hot Mix Asphalt (HMA) Design; source PDF pp. 193–280. Open publication.
- Asphalt Institute. (2007). The asphalt handbook (7th ed., MS-4), Chapter 10: Quality Control and Acceptance of Hot Mix Asphalt; source PDF pp. 479–552. Open publication.
- British Columbia Ministry of Transportation. (2006). Design build standard specifications, Section 502; PDF pp. 297–318. Historical contract exercise. Open publication.
- BC Ministry of Transportation and Transit. (2025). 2025 Standard Specifications for Highway Construction, Volume 1. Open publication.

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Study chapter · Outline G3

Application

Application rate describes how much asphalt mixture was placed over a measured area. It is a mass-per-area quantity. It can help identify a quantity or thickness concern, but does not replace density, level or compacted-thickness checks.

Define Section 502

LEARNING STEP 1

Define Section 502

Section 502.14 of the 2006 specification determines application rate from the delivered tonnage assigned to the lot and its covered area, with the specified treatment of entrance letdowns and intersections. Reconcile weigh tickets, returned material, location and lift according to the contract. Counting an entire truck twice or including a ticket from another lane changes the result even when the arithmetic is correct.

Use consistent units: 1 tonne = 1,000 kg. Actual application rate in kg/m² = accepted assigned mass in kg ÷ measured area in m². The ratio used by Table 502-H is actual rate ÷ specified rate × 100. This ratio is a percentage of the specified rate, not a binder percentage and not percent density.

The dated table distinguishes bottom/single lifts from the top lift of multiple lifts. For the uncomplicated 96.0–103.9% band, both columns show +$0.50/t. Other bands include different deductions, restrictions on payment for excess material and rejection/remedial provisions. Read the full row and correct lift column. Do not treat over-application as a desirable way to earn payment: it can change grade, tie-ins and material demand.

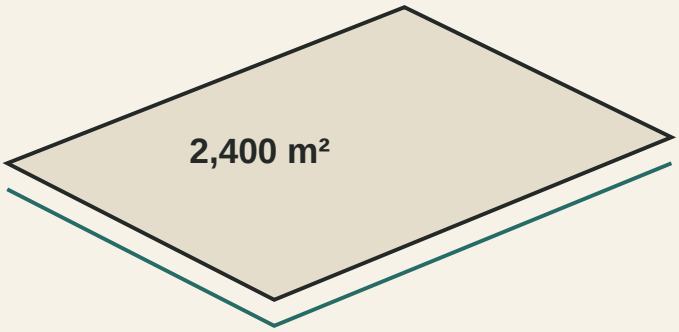
If a calculated rate seems wrong, check area boundaries, units, tickets, lift identity and authorized changes first. Then compare with thickness and survey evidence. A rate alone cannot prove a uniform mat; a thick patch and a thin patch can average to the specified rate.

Mass divided by area gives application rate

ACCESS IS4S / ASPHALT STUDY

Mass divided by area gives application rate

Fictional lot: 288 t over 2,400 m²; specified rate 120 kg/m².



CALCULATE

288 t = 288,000 kg
288,000 ÷ 2,400
= 120 kg/m²

120 ÷ 120 × 100 = 100% of the specified rate

An average rate does not prove uniform thickness or full acceptance.

Conceptual teaching diagram · Use the stated case and actual project/machine requirements.

Note. ACCESS IS4S. (2026). Mass divided by area gives application rate [Original instructional SVG]. Conceptual explanation based on the cited chapter readings; geometry and operating arrangements are illustrative, not prescribed settings.

2006 Table 502-H: application-rate adjustment

TABLE 502-H – PAYMENT ADJUSTMENTS FOR MATERIAL APPLICATION RATE

Actual Application Rate Expressed as % of Specified Application Rate	Assessment for Application Rate $ per tonne	
	Bottom Lift or Single Lift	Top Lift of Multiple Lifts
110 +	-$6.00 for all material in the Lot up to 110% and no payment for product in excess of 110.0%	No payment
106.0 – 109.9	-$4.00 for all material in the Lot	-$6.00 for all material in the Lot up to 106% and no payment for product in excess of 106.0%
105.0 – 105.9		-$4.00 for all material in the Lot
104.0 – 104.9	$0.00	-$2.00 for all material
96.0– 103.9	+$0.50	+ $0.50
94.0 - 95.9	-$1.00	-$1.00
92.0 - 93.9	-$2.00	-$2.00
90.0 - 91.9	-$3.00	-$3.00
85.0 – 89.9	-$5.00	-$5.00
84.9 or lower	Rejected, Mill and Fill, and/or rejected with no remedial work required at the discretion of the Ministry Representative	Rejected, Mill and Fill, Overlay, and/or rejected with no remedial work required at the discretion of the Ministry Representative

Note. British Columbia Ministry of Transportation. (2006). Design build standard specifications, Section 502 (section p. 14; PDF p. 310). Historical contract-reading example; complete clauses and actual project adoption govern. Full source context.

Worked example · Mass over area

A fictional lot receives 288 t over 2,400 m². Specified application is 120 kg/m².

Think it through.

288 t = 288,000 kg. Actual = 288,000/2,400 = 120 kg/m². Ratio = 120/120 × 100 = 100%. In the dated 96.0–103.9% band, isolated adjustment = 288 × $0.50 = +$144, subject to all applicable conditions.

Try it and explain your thinking

A 200 m × 3.5 m rectangle receives 77 t

A 200 m × 3.5 m rectangle receives 77 t. Find actual application rate.

Skills you use: Numeracy, Reading.

Specified rate is 115 kg/m²

Specified rate is 115 kg/m². What percentage of specified is 110 kg/m²?

Skills you use: Numeracy, Problem solving.

Design a ticket register that a partner can audit.

Design a ticket register that a partner can audit.

Skills you use: Digital, Writing, Collaboration, Creativity and innovation.

Show the skill in supervised practice

Machine/model/serial or classroom case: _____ · Manual/procedure and revision: _____ · Date/location: _____ · Learner/observer: _____

Record each criterion as 0—not yet demonstrated, 1—with task-solving support, or 2—independently under the stated conditions. Mark unobserved criteria separately; do not convert missing opportunity into a zero. The assessor determines the required repeat practice and applicable qualification requirements. A critical failure means hold, correction and reassessment regardless of the numerical result. Use safe simulations for classroom reasoning; actual operation requires the selected equipment, approved conditions and competent supervision.

What the observer checks	Result	Evidence
Critical: Safe access, surroundings and communication maintained.	Pass / hold	Observation: _____
Critical: Required energy, heat, movement and public-access controls maintained for the actual task.	Pass / hold	Observation: _____

What the observer checks	Result	Evidence
Measure and document a lot boundary.	0 / 1 / 2	Evidence/help: _____
Reconcile unique tickets to the correct lift.	0 / 1 / 2	Evidence/help: _____
Calculate rate and explain why average rate does not prove uniform thickness.	0 / 1 / 2	Evidence/help: _____

Actual conditions and support: _____

Feedback, next practice and authorized release/status: _____

Sources and further reading

- SkilledTradesBC. (2016). *Asphalt Paving/Laydown Technician program outline*, G3, printed p. 54; source PDF pp. 56–56.
- Asphalt Institute. (2007). The asphalt handbook (7th ed., MS-4), Chapter 8: Hot Mix Asphalt Paving; source PDF pp. 391–446. Open publication.
- British Columbia Ministry of Transportation. (2006). Design build standard specifications, Section 502; PDF pp. 297–318. Historical contract exercise. Open publication.
- BC Ministry of Transportation and Transit. (2025). 2025 Standard Specifications for Highway Construction, Volume 1. Open publication.

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Study chapter · Outline G4

Segregation

Segregation is separation within the mixture that produces an uneven distribution of aggregate sizes or other mixture characteristics. A visibly coarse patch can be a clue, but a crew needs agreed inspection criteria and a traceable location before deciding its cause or treatment.

Define Section 502

LEARNING STEP 1

Define Section 502

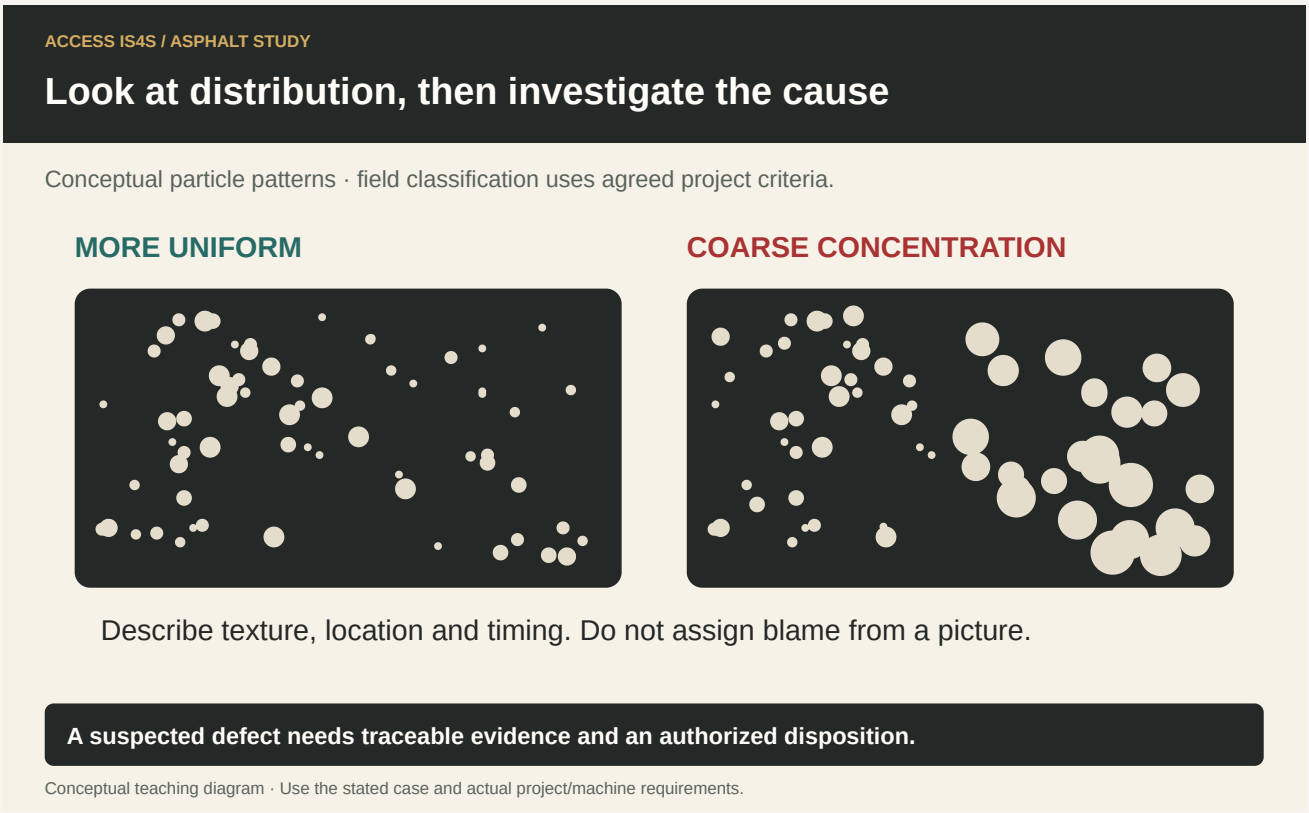
Section 502.15 of the 2006 specification calls for a homogeneous mixture and uniform surface texture. Its visual classification distinguishes slight, moderate and severe segregation using the amount of stone and remaining binder/fine-aggregate matrix. The project's agreed comparison areas or photographs support consistent judgement. Temperature differences can also affect paving behaviour; thermal segregation and aggregate-size segregation are related quality concerns but are not interchangeable descriptions.

Track where a pattern occurs: at truck changes, the end of loads, edges, behind a particular part of the screed or at repeated stops. Material handling, stockpiles, truck loading/unloading, hopper management and the feed system all deserve investigation. Do not claim a single cause from a photograph alone. Record station, lane, lift, time, load and observed texture, then notify the responsible lead while the crew can still investigate the process.

The historical top-lift tables 502-I-1 and I-2 address remediation and payment. They include different remedies by severity and counts per lane kilometre. These dated repair descriptions are an exercise in reading a contract, not a field instruction to apply slurry or overlay without approval. Current project specifications and the authorized repair design govern actual work. A payment deduction does not automatically remove the requirement for remediation.

Prevention is a coordinated task. Maintain the specified material-handling and paving process, communicate supply changes early, and avoid scattering coarse material onto a finished surface. When a defect is suspected, preserve evidence and obtain direction before an improvised repair hides the cause.

Look at distribution, then investigate the cause



Note. ACCESS IS4S. (2026). Look at distribution, then investigate the cause [Original instructional SVG]. Conceptual explanation based on the cited chapter readings; geometry and operating arrangements are illustrative, not prescribed settings.

Segregation along the outside edge of a mat



Figure 8.26 Segregation Along Outside Edge of Mat

Note. Asphalt Institute. (2007). The asphalt handbook (7th ed., MS-4, Fig. 8.26, p. 402; PDF p. 436). Faithful scan crop; original figure caption retained. Original image contributor: James A. Scherocman, P.E., Consulting Engineer; credit register, PDF pp. 803–805. Full source context.

Worked example · Describe before diagnosing

A coarse-looking strip appears near the right edge after three truck changes. The surrounding surface has more fine matrix.

Think it through.

Record the repeated location and timing, photograph from a safe position with a scale and location label, and notify the paving/quality lead. Compare with agreed criteria and inspect the process under authorized conditions. Do not declare severe segregation or choose a repair from appearance alone.

Try it and explain your thinking

Write three useful observations and one unsupported conclusion for the case.

Write three useful observations and one unsupported conclusion for the case.

Skills you use: Writing, Problem solving.

In the dated Table 502-I-2, a 1 lane-km top lift has 4 slight areas, zero moderate and zero severe

In the dated Table 502-I-2, a 1 lane-km top lift has 4 slight areas, zero moderate and zero severe. What is the isolated adjustment?

Skills you use: Numeracy, Reading.

Propose a safe digital defect log and a respectful crew discussion.

Propose a safe digital defect log and a respectful crew discussion.

Skills you use: Digital, Communication, Collaboration, Creativity and innovation.

Show the skill in supervised practice

Machine/model/serial or classroom case: _____ · Manual/procedure and revision: _____ · Date/location: _____ · Learner/observer: _____

Record each criterion as 0—not yet demonstrated, 1—with task-solving support, or 2—independently under the stated conditions. Mark unobserved criteria separately; do not convert missing opportunity into a zero. The assessor determines the required repeat practice and applicable qualification requirements. A critical failure means hold, correction and reassessment regardless of the numerical result. Use safe simulations for classroom reasoning; actual operation requires the selected equipment, approved conditions and competent supervision.

What the observer checks	Result	Evidence
Critical: Safe access, surroundings and communication maintained.	Pass / hold	Observation: _____
Critical: Required energy, heat, movement and public-access controls maintained for the actual task.	Pass / hold	Observation: _____
Compare texture using approved project criteria.	0 / 1 / 2	Evidence/help: _____
Locate and record a suspected segregated area.	0 / 1 / 2	Evidence/help: _____
Separate observation, possible cause and authorized disposition.	0 / 1 / 2	Evidence/help: _____

Actual conditions and support: _____

Feedback, next practice and authorized release/status: _____

Sources and further reading

- SkilledTradesBC. (2016). *Asphalt Paving/Laydown Technician program outline*, G4, printed p. 55; source PDF pp. 57–57.
- Asphalt Institute. (2007). The asphalt handbook (7th ed., MS-4), Chapter 8: Hot Mix Asphalt Paving; source PDF pp. 391–446. Open publication.
- Caterpillar. (2016). Guide to Asphalt Paving (QEBQ1888). Asphalt paving process, tractor, screed, material delivery and handwork; model examples vary. Open publication.
- British Columbia Ministry of Transportation. (2006). Design build standard specifications, Section 502; PDF pp. 297–318. Historical contract exercise. Open publication.
- BC Ministry of Transportation and Transit. (2025). 2025 Standard Specifications for Highway Construction, Volume 1. Open publication.

Figure-specific pages, editions and original contributors are credited beside each image. Current project requirements and the matching machine instructions determine practical use.

Density

Percent density compares measured in-place density with a defined reference density. The reference is part of the result. Saying only “the density is 94%” leaves out information needed to interpret it.

Define Section 502

LEARNING STEP 1

Define Section 502

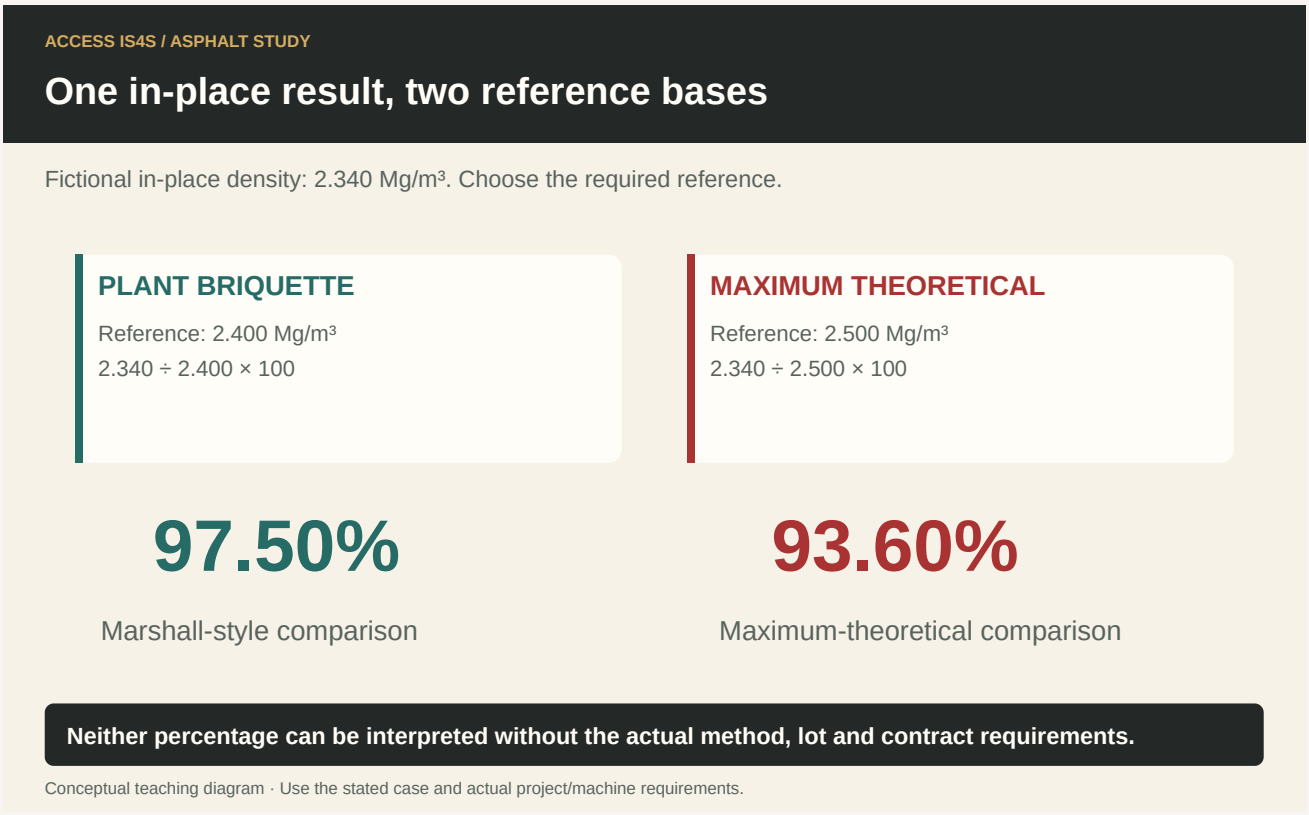
Section 502.11 of the supplied 2006 document uses different references for Marshall and Superpave mixtures. The Marshall expression divides in-place density by the density of the plant briquette. The Superpave expression divides in-place density by maximum theoretical density determined by the specified procedure. Use compatible units, the applicable method and representative results from the correct lot.

A core and a nuclear gauge do not produce evidence through the same procedure. Follow the approved sampling, testing, calibration/correlation and reporting requirements. Nuclear gauge operation requires the applicable CNSC licensing, training and controls; reading this chapter does not authorize it. Record location, lift, sample/gauge identification, method, density units and reference result. Avoid substituting a different day’s reference because it produces a more favourable percentage.

The dated tables E-1 and E-2 use different acceptance/payment bands. For example, a Marshall lot average of 97.6–98.0% has a +$0.50/t adjustment, while the Superpave +$0.50/t band is 93.0–93.9%. These bands cannot be exchanged. Sub-lot limits, rejection provisions and the other acceptance requirements also apply. A satisfactory lot mean can conceal an unacceptable individual area.

Compaction effort, temperature, mixture behaviour, support and rolling pattern affect the outcome. More passes are not automatically better: they may be ineffective after cooling or damage an unstable mat. Use the approved pattern and quality feedback to guide authorized changes.

One in-place result, two reference bases



Note. ACCESS IS4S. (2026). One in-place result, two reference bases [Original instructional SVG]. Conceptual explanation based on the cited chapter readings; geometry and operating arrangements are illustrative, not prescribed settings.

2006 Tables 502-E-1 and E-2: different density bases

TABLE 502-E-1 – PAYMENT ADJUSTMENTS FOR DENSITY – MARSHALL

% Density Lot Average	Payment Adjustment ($ Per Tonne) for Design Lift Thickness
98.6 Plus	+$1.00
98.1 – 98.5	+$0.75
97.6 – 98.0	+$0.50
97.0 – 97.5	+$0.25
96.6 – 96.9	-$0.50
96.0 – 96.5	-$1.00
Less than 96.0%	Reject

If any Sub-Lot has a density below 96%, the Sub-Lot shall be considered rejectable, and, subject to the Contractor's opportunity to increase the density as described in DBSS 502.11.04, shall be rejected.

TABLE 502-E-2 – PAYMENT ADJUSTMENTS FOR DENSITY - SUPERPAVE

% Density Lot Average	Payment Adjustment ($ Per Tonne) for Design Lift Thickness
95.0 Plus	+$1.00
94.0 – 94.9	+$0.75
93.0 – 93.9	+$0.50
92.0 – 92.9	+$0.25
91.6 – 91.9	-$0.50
91.0 – 91.5	-$1.00
less than 91.0 %	Reject

If any Sub-Lot has a density below 91%, the Sub-Lot shall be considered rejectable, and, subject to the Contractor's opportunity to increase the density as described in DBSS 502.11.04, shall be rejected. Percent Density is based on a comparison to the Maximum Theoretical Density.

Note. British Columbia Ministry of Transportation. (2006). Design build standard specifications, Section 502 (section p. 11; PDF p. 307). Historical contract-reading example; complete clauses and actual project adoption govern. Full source context.

Worked example · Two legitimate references, two different percentages

In-place density is 2.340 Mg/m³. A fictional plant briquette result is 2.400 Mg/m³; maximum theoretical density is 2.500 Mg/m³.

Think it through.

Marshall-style comparison: 2.340/2.400 × 100 = 97.50%. Maximum-theoretical comparison: 2.340/2.500 × 100 = 93.60%. Both calculations are arithmetic examples; use only the reference required for the actual mixture and contract.

Try it and explain your thinking

For 2.325 Mg/m³ in place and 2.500 Mg/m³ maximum theoretical, calculate percent.

For 2.325 Mg/m³ in place and 2.500 Mg/m³ maximum theoretical, calculate percent.

Skills you use: Numeracy, Reading.

A 400 t fictional Superpave lot average is 93.4%

A 400 t fictional Superpave lot average is 93.4%. Find the isolated dated table adjustment, then name two checks still needed.

Skills you use: Numeracy, Problem solving.

A spreadsheet divides every test by a fixed 2.400

A spreadsheet divides every test by a fixed 2.400. What should a reviewer do?

Skills you use: Digital, Collaboration, Writing.

Show the skill in supervised practice

Machine/model/serial or classroom case: _____ · Manual/procedure and revision: _____ · Date/location: _____ · Learner/observer: _____

Record each criterion as 0—not yet demonstrated, 1—with task-solving support, or 2—independently under the stated conditions. Mark unobserved criteria separately; do not convert missing opportunity into a zero. The assessor determines the required repeat practice and applicable qualification requirements. A critical failure means hold, correction and reassessment regardless of the numerical result. Use safe simulations for classroom reasoning; actual operation requires the selected equipment, approved conditions and competent supervision.

What the observer checks	Result	Evidence
Critical: Safe access, surroundings and communication maintained.	Pass / hold	Observation: _____
Critical: Required energy, heat, movement and public-access controls maintained for the actual task.	Pass / hold	Observation: _____
Explain and select the correct density denominator.	0 / 1 / 2	Evidence/help: _____
Trace test results to location and reference.	0 / 1 / 2	Evidence/help: _____
Distinguish a lot average from individual acceptance requirements.	0 / 1 / 2	Evidence/help: _____

Actual conditions and support: _____

Feedback, next practice and authorized release/status: _____

Sources and further reading

- SkilledTradesBC. (2016). *Asphalt Paving/Laydown Technician program outline*, G5, printed p. 56; source PDF pp. 58–58.
- Asphalt Institute. (2007). The asphalt handbook (7th ed., MS-4), Chapter 9: Hot Mix Asphalt Compaction; source PDF pp. 447–478. Open publication.
- Asphalt Institute. (2007). The asphalt handbook (7th ed., MS-4), Chapter 10: Quality Control and Acceptance of Hot Mix Asphalt; source PDF pp. 479–552. Open publication.
- British Columbia Ministry of Transportation. (2006). Design build standard specifications, Section 502; PDF pp. 297–318. Historical contract exercise. Open publication.
- Troxler Electronic Laboratories. (Edition 1.3, February 2021). RoadReader Model 3430 Surface Moisture-Density Gauge: Manual of Operation and Instruction. Open publication.
- Canadian Nuclear Safety Commission. (Web snapshot: October 30, 2026). Portable Nuclear Gauges. Open publication.

Figure-specific pages, editions and original contributors are credited beside each image. Current project requirements and the matching machine instructions determine practical use.

Study chapter · Outline G6

Smoothness

Smoothness concerns the longitudinal profile experienced by a moving vehicle. It differs from crossfall, compacted thickness and density. A mat can meet one of those requirements while failing another.

Define Section 502

LEARNING STEP 1

Define Section 502

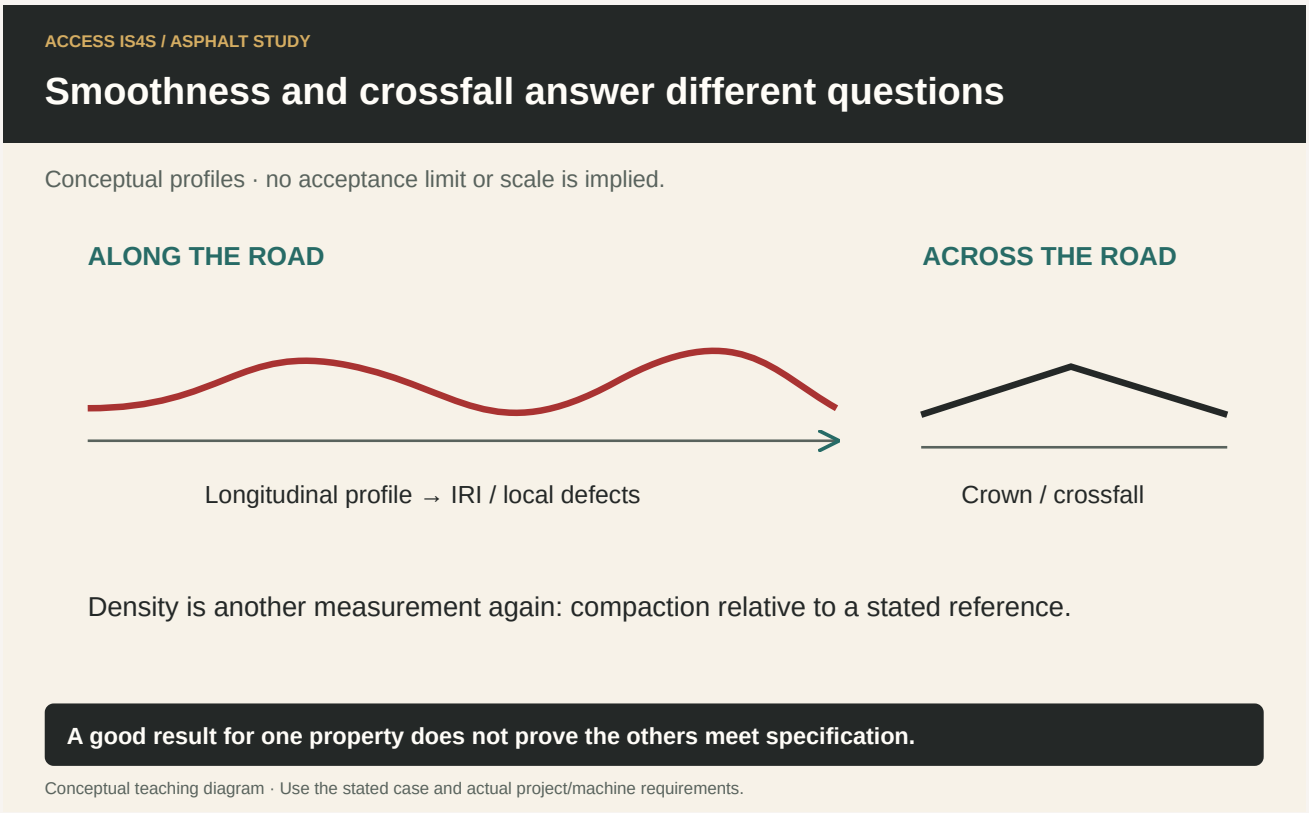
Section 502.16 of the dated 2006 specification measures longitudinal profile with a specified precision rolling instrument and computes the International Roughness Index (IRI). IRI is commonly expressed in m/km; lower values indicate a smoother profile under the model. It is not a direct measurement of crossfall or a single bump's height. Straightedge checks and localized-defect requirements answer different questions and may apply in addition to IRI.

Identify the lane, direction, station limits, lift and applicable lot/sub-lot rules. The historical clauses treat auxiliary lanes and short lengths specifically, and use the applicable Table 502-J or J-2 when adopted. Acceptance limits and payment schedules depend on the road classification and contract. Do not import an IRI threshold from another job. Preserve the instrument record and the quality lead's interpretation.

Smoothness begins before the profiler arrives: prepared base, correct setup, steady paving speed and material head, coordinated truck changes, careful joints and suitable rolling practice all matter. Abrupt control changes, stops and poorly matched joints can leave profile defects. A roller cannot reliably erase every paver setup error. Report an emerging pattern early enough for the lead to investigate the process.

The 2006 auxiliary-lane example prorates a qualifying adjustment by segment length relative to a 1 km lot. This is a contract rule, not a general conversion from IRI to dollars. Repairs and remeasurement require authorized procedures. Keep the original and subsequent profiles linked so the correction can be evaluated transparently.

Smoothness and crossfall answer different questions



Note. ACCESS IS4S. (2026). Smoothness and crossfall answer different questions [Original instructional SVG]. Conceptual explanation based on the cited chapter readings; geometry and operating arrangements are illustrative, not prescribed settings.

2006 Tables 502-J and J-2: smoothness

TABLE 502-J – LOT ASSESSMENT AND PAYMENT ADJUSTMENTS FOR SMOOTHNESS

For the final surface course only, the following payment adjustments shall apply to each Lot.

Lot IRI (m/km)	Payment Adjustment
≤ 0.80	+$2,000
0.81 to 0.90	+$1000
0.91 to 1.00	+$500
1.01 to 1.10	+$200
1.11 to 1.20	0
1.21 to 1.30	-$100
1.31 to 1.40	-$250
1.41 to 1.50	-$600
1.51 to 1.60	-$1,400
1.61 to 1.70	-$2,000
1.71 to 1.80	-$3,000
1.80	**ACCEPTANCE LIMIT.** Reject or corrective work required above this level

The Lot IRI is the average IRI value calculated for the Sub-Lots within the Lot.

TABLE 502-J-2 – LOT ASSESSMENT AND ALTERNATIVE PAYMENT ADJUSTMENTS FOR SMOOTHNESS

For the final surface course only, and only where specifically invoked by the Contract, the following payment adjustments shall apply to each Lot on a road designated by the Ministry for application of alternative smoothness payment adjustments

Lot IRI (m/km)	Payment Adjustment
<1.00	+$2000
1.01 – 1.10	+$1000
1.11 – 1.20	+$500
1.21 – 1.30	+$250
1.31 – 1.40	0
1.41 – 1.60	-$300
1.61 – 1.80	-$750
1.81 – 1.90	-$1500
1.90 – 2.00	-$2500
2.00	**ACCEPTANCE LIMIT.** Reject or corrective work required above this level

The Lot IRI is the average IRI calculated for the Sub-Lots within the Lot

Note. British Columbia Ministry of Transportation. (2006). Design build standard specifications, Section 502 (section p. 16; PDF p. 312). Historical contract-reading example; complete clauses and actual project adoption govern. Full source context.

Worked example · Prorating an authorized historical adjustment

For a fictional exercise, the applicable table and acceptance checks establish a +$800 adjustment for a full 1 km lot. The qualifying auxiliary segment is 565 m.

Think it through.

Length factor = 565/1,000 = 0.565. Prorated adjustment = 0.565 × $800 = +$452. The $800 is an explicitly supplied exercise value, not a quoted Table 502-J value or a current project entitlement.

Try it and explain your thinking

A qualifying segment is 240 m and the supplied full-kilometre adjustment is −$1,200

A qualifying segment is 240 m and the supplied full-kilometre adjustment is −$1,200. Calculate the exercise adjustment.

Skills you use: Numeracy, Reading.

A learner says a good density result proves a smooth road

A learner says a good density result proves a smooth road. Explain the gap.

Skills you use: Communication, Problem solving.

Create a handover for a repeated bump near each truck-change location.

Create a handover for a repeated bump near each truck-change location.

Skills you use: Writing, Digital, Collaboration, Adaptability.

Show the skill in supervised practice

Machine/model/serial or classroom case: _____ · Manual/procedure and revision: _____ · Date/location: _____ · Learner/observer: _____

Record each criterion as 0—not yet demonstrated, 1—with task-solving support, or 2—independently under the stated conditions. Mark unobserved criteria separately; do not convert missing opportunity into a zero. The assessor determines the required repeat practice and applicable qualification requirements. A critical failure means hold, correction and reassessment regardless of the numerical result. Use safe simulations for classroom reasoning; actual operation requires the selected equipment, approved conditions and competent supervision.

What the observer checks	Result	Evidence
Critical: Safe access, surroundings and communication maintained.	Pass / hold	Observation: _____

What the observer checks	Result	Evidence
Critical: Required energy, heat, movement and public-access controls maintained for the actual task.	Pass / hold	Observation: _____
Identify the profile measurement and applicable contract limits.	0 / 1 / 2	Evidence/help: _____
Separate IRI, localized defects and crossfall.	0 / 1 / 2	Evidence/help: _____
Trace a profile concern to a location and authorized corrective process.	0 / 1 / 2	Evidence/help: _____

Actual conditions and support: _____

Feedback, next practice and authorized release/status: _____

Sources and further reading

- SkilledTradesBC. (2016). *Asphalt Paving/Laydown Technician program outline*, G6, printed p. 57; source PDF pp. 59–59.
- Asphalt Institute. (2007). The asphalt handbook (7th ed., MS-4), Chapter 10: Quality Control and Acceptance of Hot Mix Asphalt; source PDF pp. 479–552. Open publication.
- British Columbia Ministry of Transportation. (2006). Design build standard specifications, Section 502; PDF pp. 297–318. Historical contract exercise. Open publication.
- BC Ministry of Transportation and Transit. (2025). 2025 Standard Specifications for Highway Construction, Volume 1. Open publication.

Figure-specific pages, editions and original contributors are credited beside each image. Current project requirements and the matching machine instructions determine practical use.

Integrated practice · all nine Skills for Success

Check the grade and the quantity

> This is a complete fictional classroom case. It does not describe an actual community, Nation, employer or project. Learners may use the supplied case without disclosing personal or cultural knowledge.

A fictional rectangle is 80 m × 3.5 m. Assigned mix mass is 33.6 t and specified application rate is 120 kg/m². Benchmark EL is 100.000 m, backsight 1.300 m, foresight 1.720 m and target base EL 99.650 m. A draft claims “the application rate proves the whole mat is accepted.” No density or smoothness results are supplied.

Reading · Check the facts

Identify the controlling facts, missing information and one unsupported statement in the brief.

Writing · Make the handover useful

Draft a concise handover with location, quantities, status and the next responsible action.

Numeracy · Calculate and check

Show the case calculation with units and an independent reasonableness check.

Digital · Direct AI and verify the files

Use an available AI assistant to draft a findings table and editable record from this brief. Request DOCX or a structured document, XLSX with formulas where useful, and a PDF/HTML reading copy. Audit every claim and reopen the actual outputs.

Problem solving · Investigate the discrepancy

Explain the discrepancy, what evidence is needed and who should authorize the next step.

Communication · Send and check the message

Brief a partner, ask for a check-back and repair one misunderstanding if it occurs.

Collaboration · Review together

Agree reviewer/author roles, complete a review and record each person’s contribution.

Adaptability · Respond to the change

Apply the change card, identify what changes and what stays, and revise the record.

Creativity and innovation · Make understanding easier

Create a labelled SVG, annotated plan, short audio explanation or captioned storyboard that makes the issue easier to understand. Test it with a reader/listener and improve one useful detail.

Change card · Use after the first attempt

A corrected drawing changes width to 3.6 m before placement; the mass allocation has not been revised.

Individual evidence record

Keep the brief, calculation, saved work, review findings and revision. Record each person's actual contribution and support. Use Not observed, Not yet demonstrated, Demonstrated with support, or Demonstrated independently for the stated action; these are local feedback descriptions.

For AI/media work, a transcript, script or storyboard is distinct from a generated audio/video file. Name what was actually produced and inspected. Do not upload private records or community-held material without the relevant authority.